

# Grid Features Defense Packet -- XB2IQX

Every trace in this packet is SIMULATED output from a fitted conductance-based single-compartment cell model (src/singlecell\_model\_v1.py), integrated deterministically at a fixed timestep -- not a raw current-clamp recording. What's being defended here is the feature-extraction methodology applied on top of that model (spike detection, burst/tonic classification, rebound detection, sag depth, adaptation ratio) -- i.e. do these algorithms correctly characterize the dynamics the model actually produces. Every panel below reruns the exact production function on a real simulated trace and marks what it found, so the classification isn't just a text label to take on faith.

cell\_floor = -4.50 nA | n grid points = 2500 | burstiness index = 0.042 | F-I slope = 19.91 Hz/nA ( $R^2=0.976$ )

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### 1. Representative traces

One trace per distinct pattern/rebound combination actually observed in this cell's grid.

### 2. Spike detection

Every detected peak overlaid on real traces.

### 3. Prominence-threshold sensitivity

How stable spike counts are across a range of thresholds.

### 4. dV/dt cross-validation

Independent detector cross-check across a random sample.

### 5. Tonic/bursting/silent classification

ISI sequence + log-ISI KDE evidence.

### 6. Rebound detection

Recovery-window rebound spikes and the latency cutoff.

### 7. Sag depth

Baseline, search window, and trough on a real trace.

### 8. Adaptation ratio

First/last ISI windows on a real, sustained tonic trace.

### 9. Firing rate / F-I slope

Rate vs. injected current with the fit line.

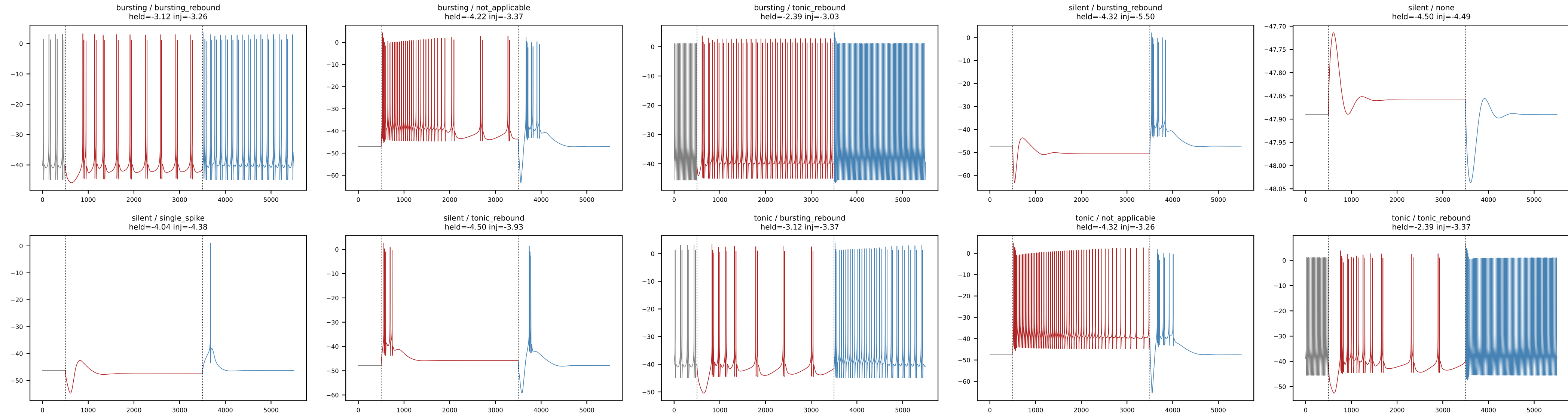
### 10. Self-consistency check

Resimulated vs. cached classification match rate.

### 11. Burst-onset-then-silence detection

Onset-burst detection and the adaptation-ratio ceased-firing gate, contrasted against a burst-onset that keeps firing.

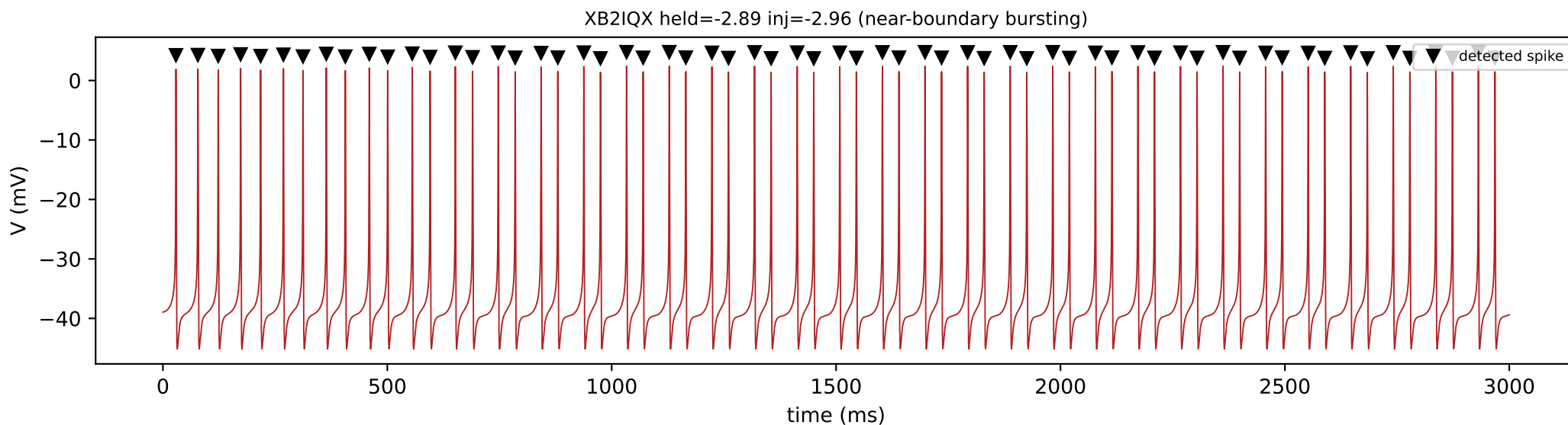
# 1. Representative traces



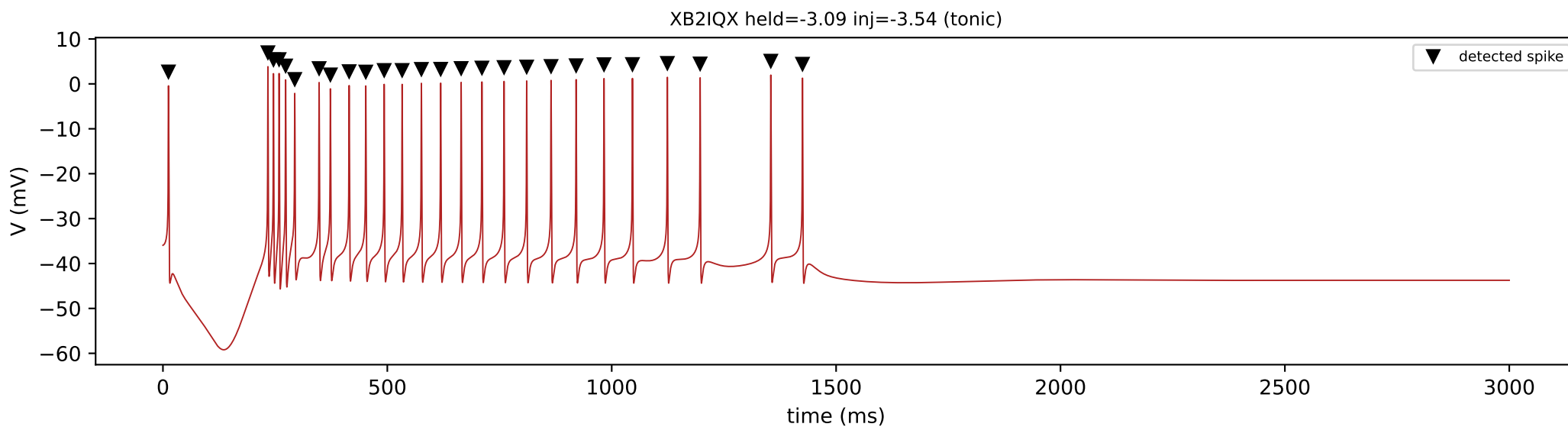
One representative trace per distinct (test-window pattern, rebound pattern) combination actually present across this cell's full 2500-point held x injected grid: gray = held baseline, red = test window (injected current), blue = recovery window (released back to held). Chosen by the same selection `plot_example_traces.py` uses for its own per-combination figures (most-prominent point, see that script's `_prominence_score`) -- this is what the cell actually does at each combination, not a hand-picked illustration.

```
python src/generate_defense_packet.py --cells XB2IQX
source: cell_held_injected_grid.pkl -> plot_example_traces.select_exemplar_points() + resimulate_point()
```

## 2. Spike detection

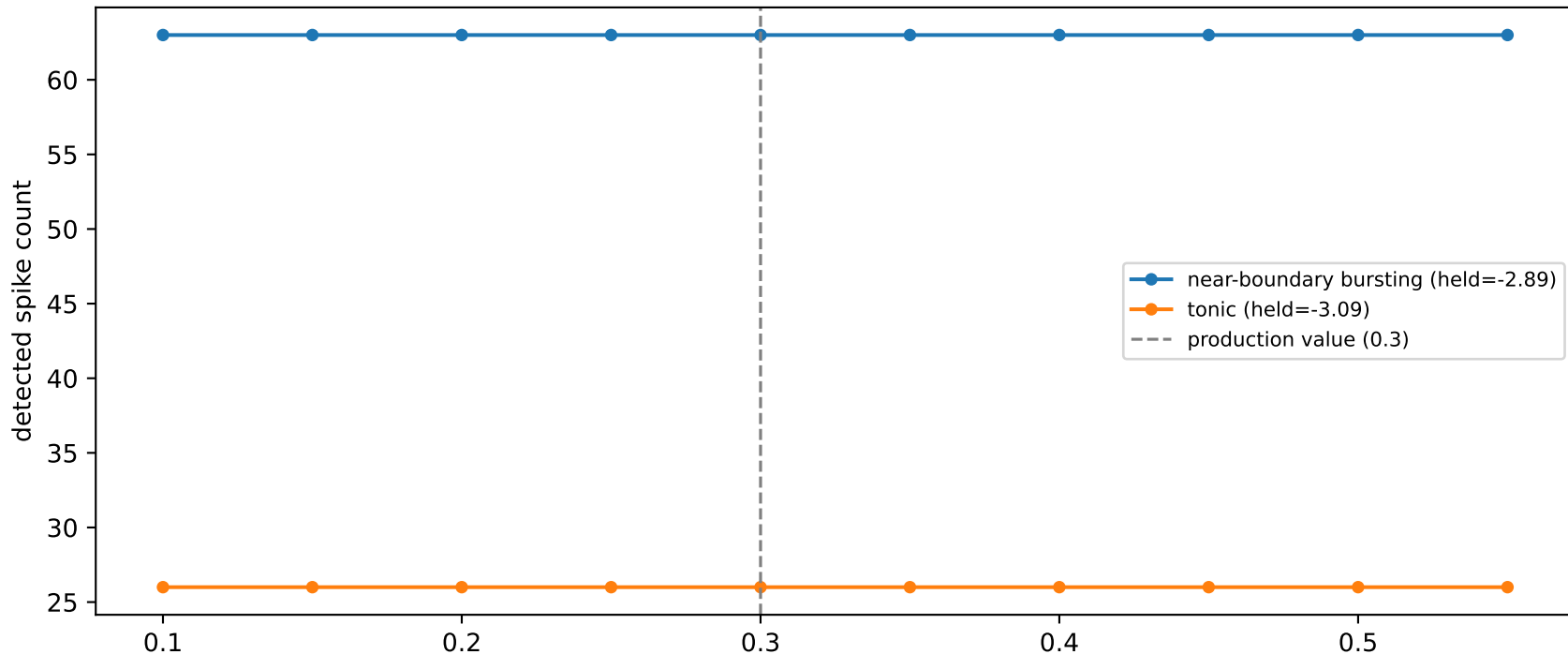


Markers show every peak `scipy.signal.find_peaks` detects at prominence  $\geq 15.0$  mV (the 15mV absolute floor, since 0.30x this trace's own 47.6mV range would be smaller) -- the exact call `count_spikes_and_rate/compute_isi_ms` use for every spike count, ISI, and burst/tonic classification in the pipeline. 63 spike(s) detected in this window.



### 3. Prominence threshold sensitivity

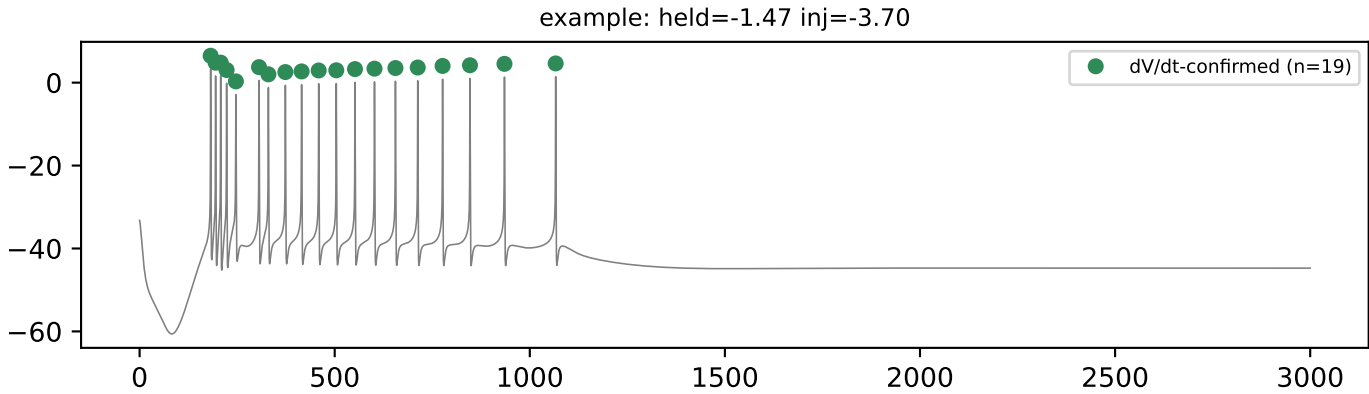
XB2IQX -- spike count vs. prominence threshold



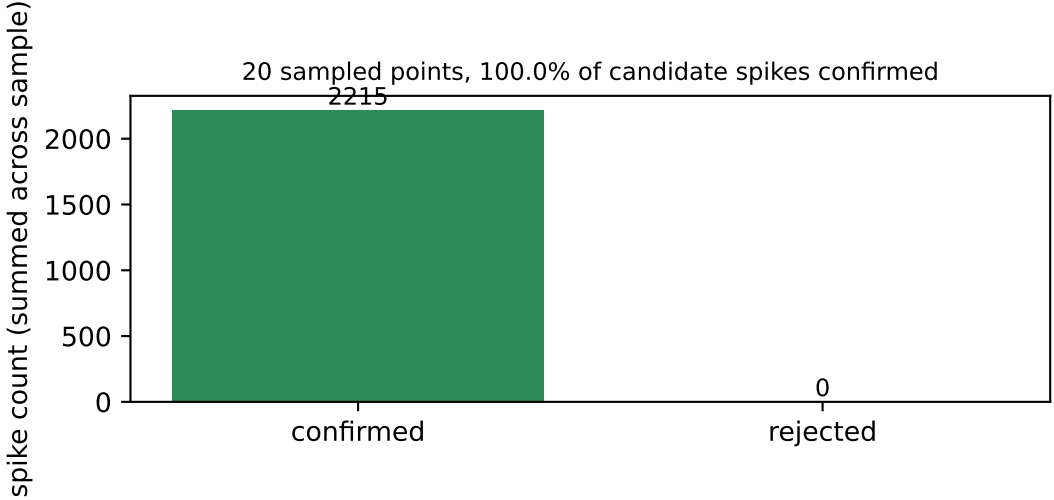
PROMINENCE\_FRACTION=0.3 is the single most meaningful, least independently-calibrated constant in the pipeline -- every spike count, ISI, and burst/tonic call across the codebase traces back to one `find_peaks(..., prominence=max(range*0.3, 15mV))` call (the absolute floor added 2026-08-13, see section 1's representative traces for why). This sweeps the fraction from 0.10 to 0.55 on real, large-amplitude spiking traces (the floor rarely binds here, since these traces' own range x0.3 already exceeds it) -- a flat plateau around the production value means the spike count this pipeline reports is not sensitive to the exact fraction chosen, not an arbitrary cliff-edge pick.

```
python src/generate_defense_packet.py --cells XB2IQX
source: cell_held_injected_grid.pkl -> resimulate_point() -> scipy.signal.find_peaks swept over PROMINENCE_FRACTION
```

#### 4. dV/dt cross-validation (unused-in-production detector as a sanity check)



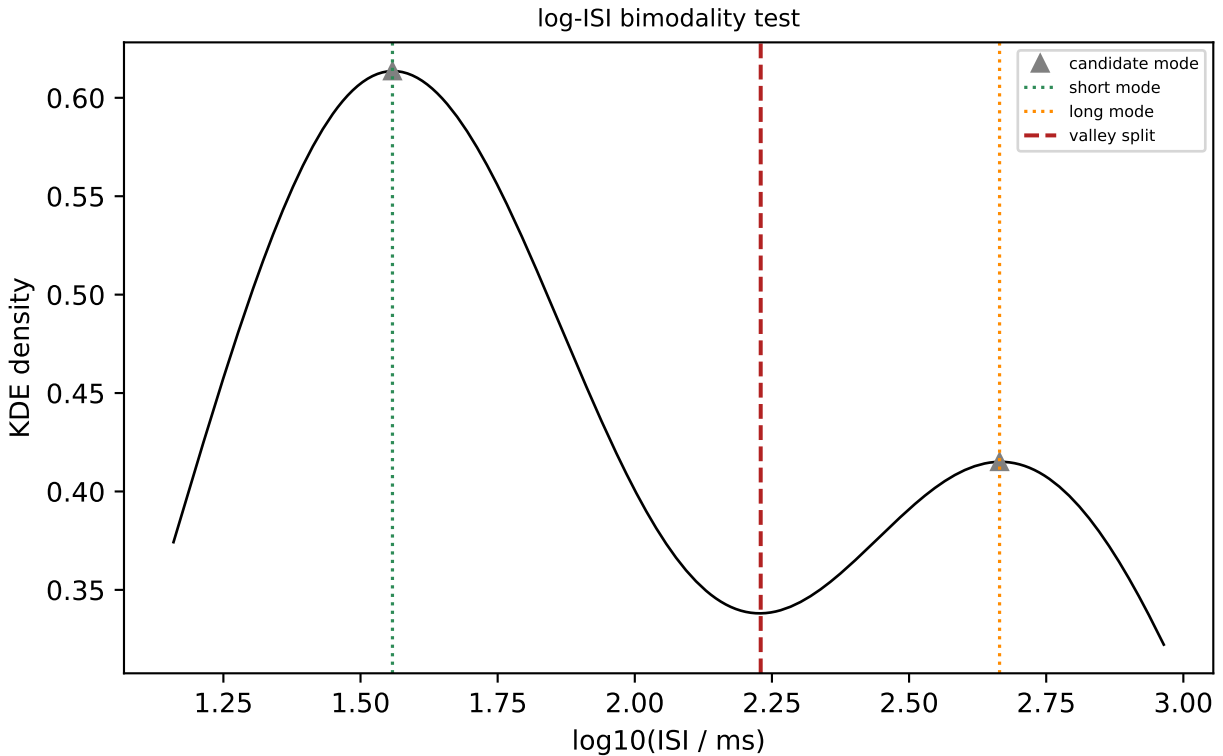
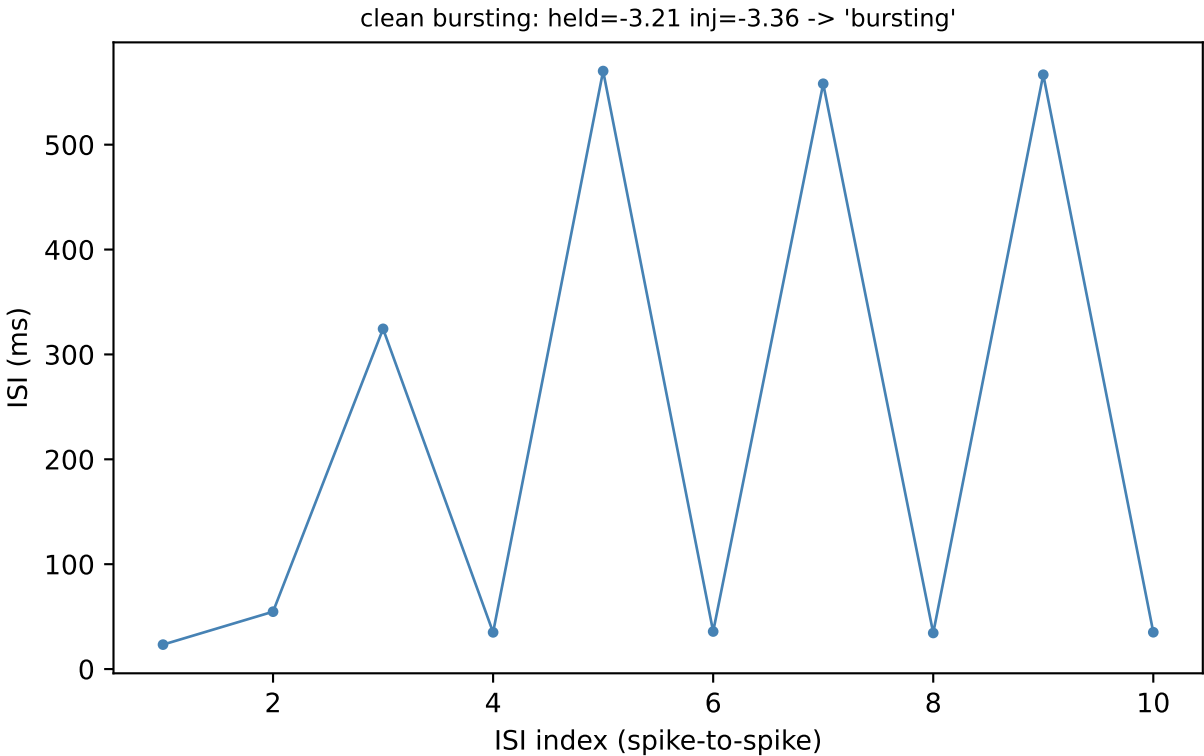
Cross-check against a second, unused-in-production detector: a candidate peak additionally requires a genuine fast upstroke ( $dV/dt \geq 10$  mV/ms within 20 ms before the peak, Bean 2007 convention) before counting. 19 of 19 candidate peak(s) confirmed; the two detectors agree on every spike in this window.



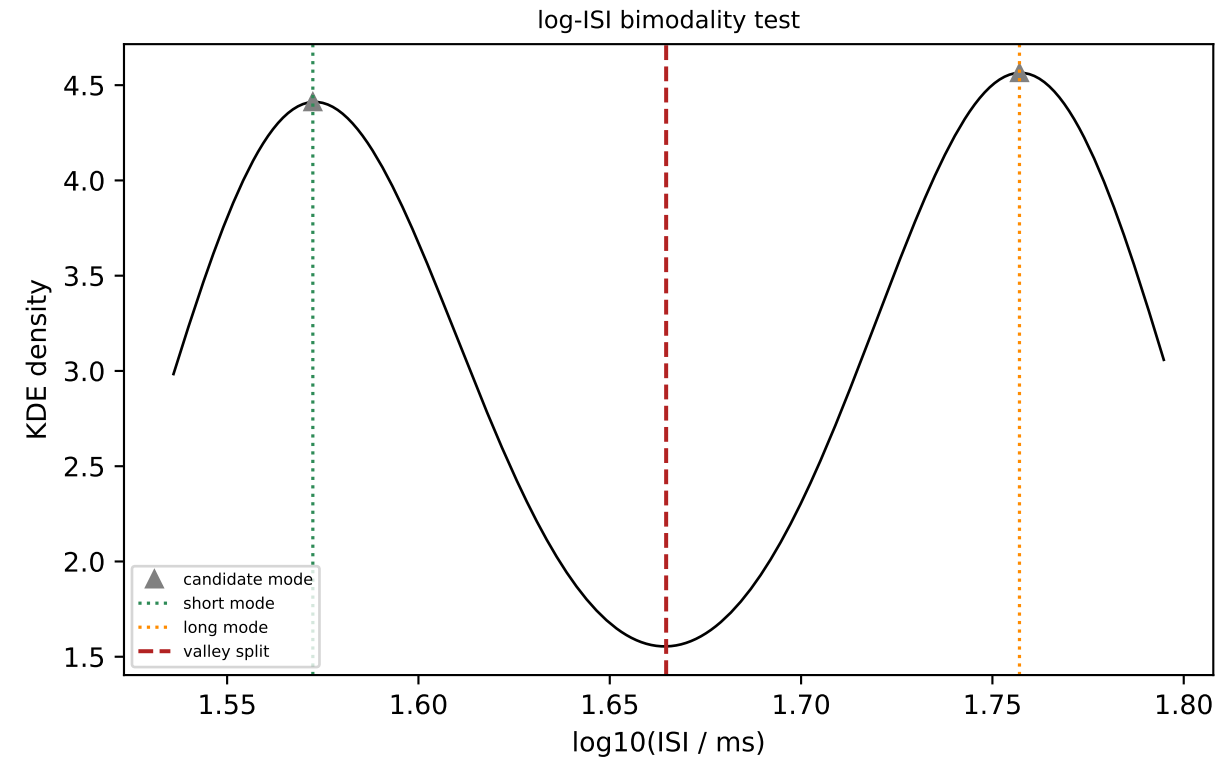
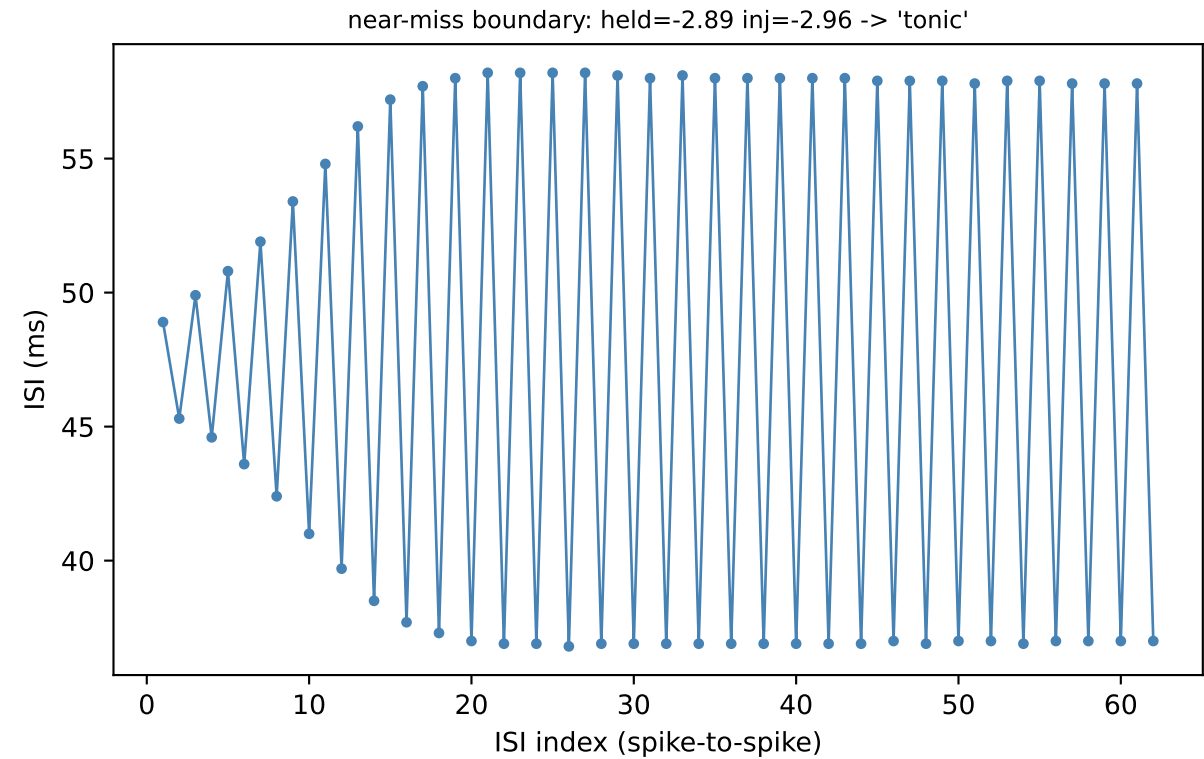
Cross-check across a random sample of 20 XB2IQX grid points: every candidate prominence-based peak is additionally required to pass a dV/dt shape-confirmation gate (detect\_spikes\_dvdt\_confirmed -- present in the codebase but not called by the production pipeline). Agreement close to 100% means the simpler production detector isn't missing a shape check that changes the answer.

```
python src/generate_defense_packet.py --cells XB2IQX
source: cell_held_injected_grid.pkl -> resimulate_point() -> find_silencing_threshold.detect_spikes_dvdt_confirmed()
```

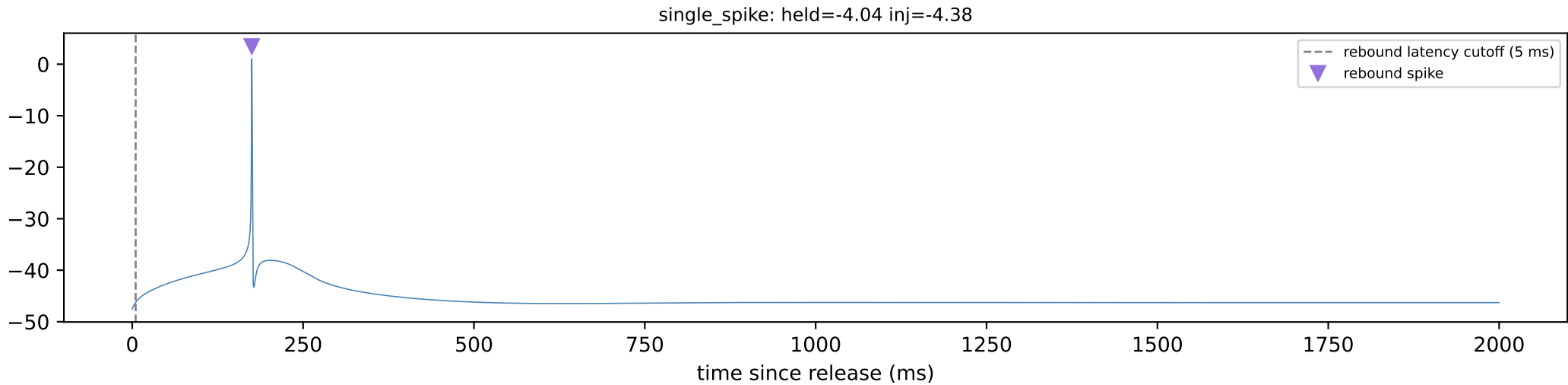
5. Tonic / bursting / silent classification



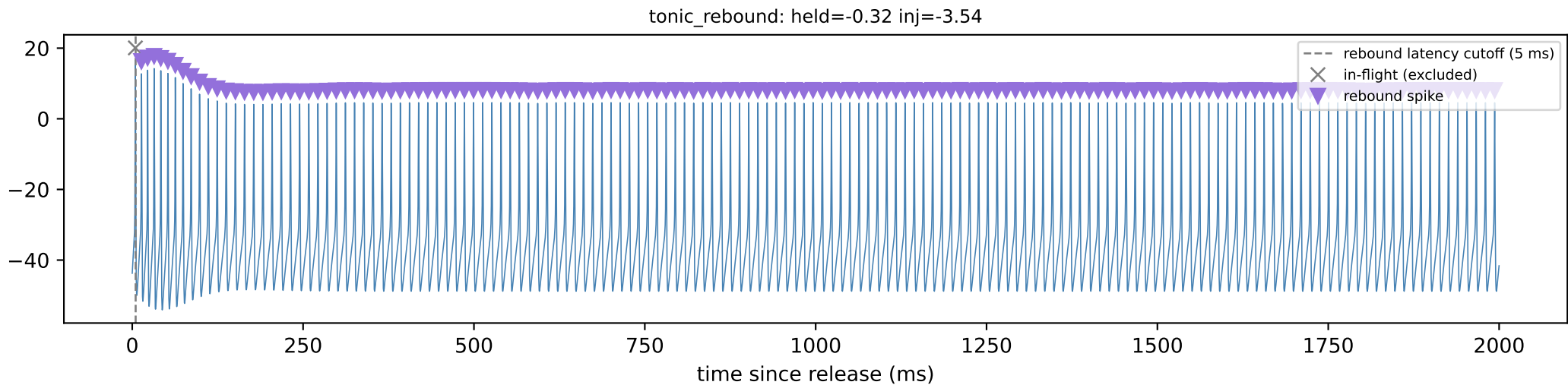
11 spikes, 10 ISIs. Split at 2 candidate mode(s) -> short  
ISI=36.4 ms, long ISI=504.8 ms (ratio 13.88x, needs >= 1.5x). Avg spikes/burst = 2.20 (needs >= 1.5). Final call:  
'bursting' (bimodality metric / Ashman's D = 6.33).



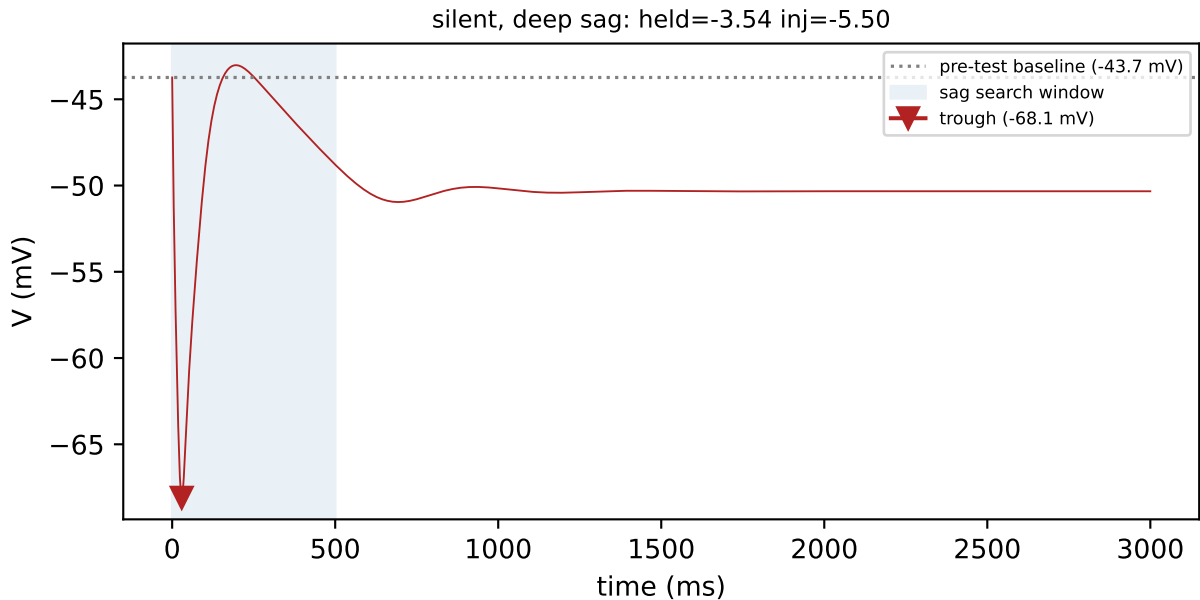
## 6. Rebound (post-inhibitory) detection



1 spike(s) at or after the 5 ms cutoff count as rebound. rebound\_occurred=True.



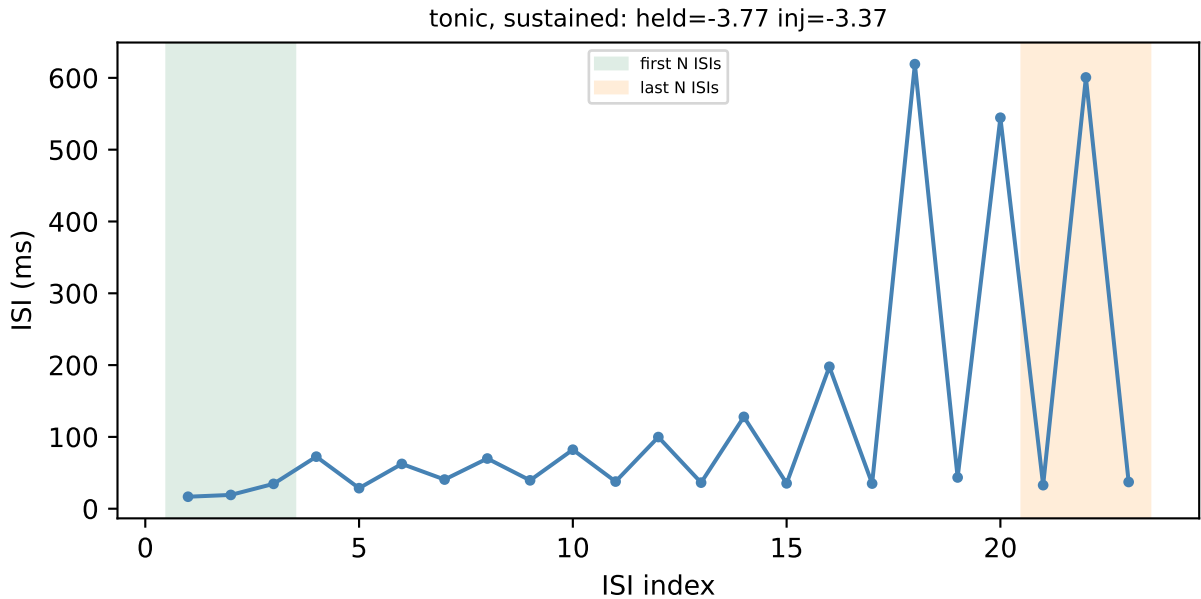
## 7. Sag depth



Sag depth = baseline - trough =  $-43.7 - (-68.1) = 24.4$  mV. Baseline is the held-current settle's final voltage; trough is the minimum voltage over the full 500 ms window (no spike before it).



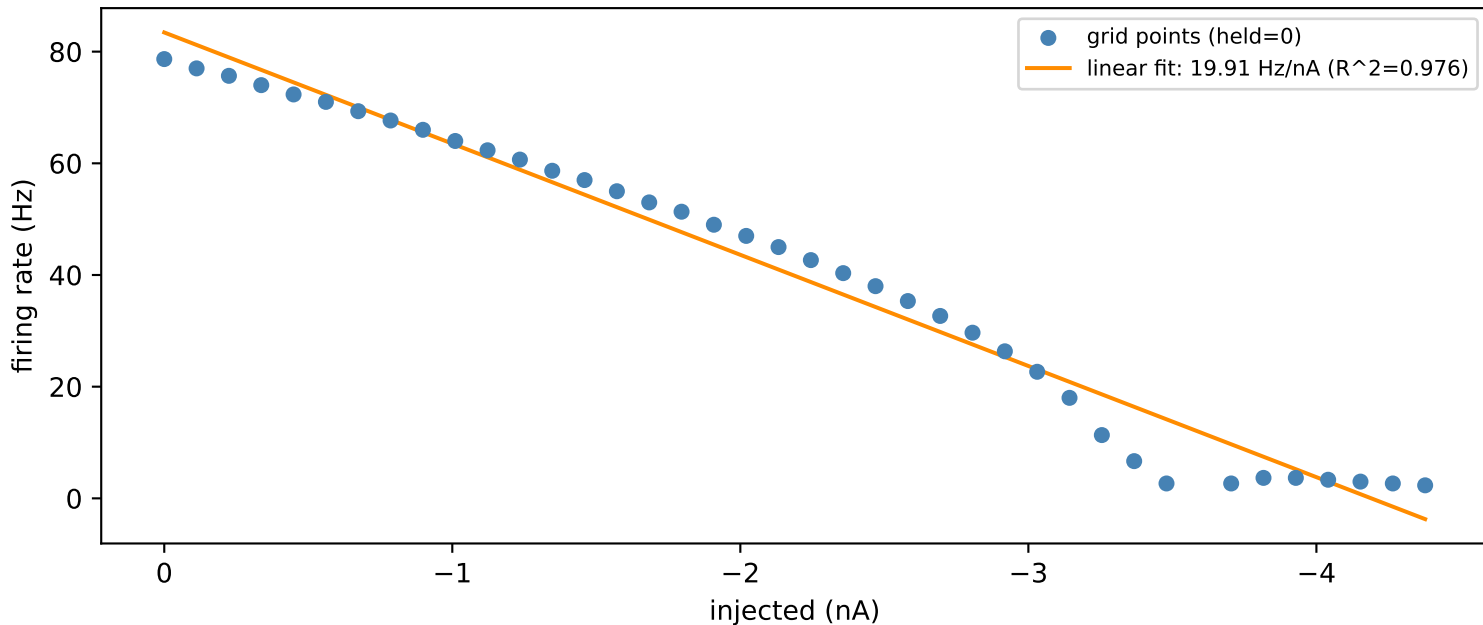
## 8. Adaptation ratio



Adaptation ratio =  $\text{mean}(\text{last 3 ISIs}) / \text{mean}(\text{first 3 ISIs}) = 9.51$ . Firing slows (adapts) over the course of this test window.

## 9. Firing rate / F-I slope

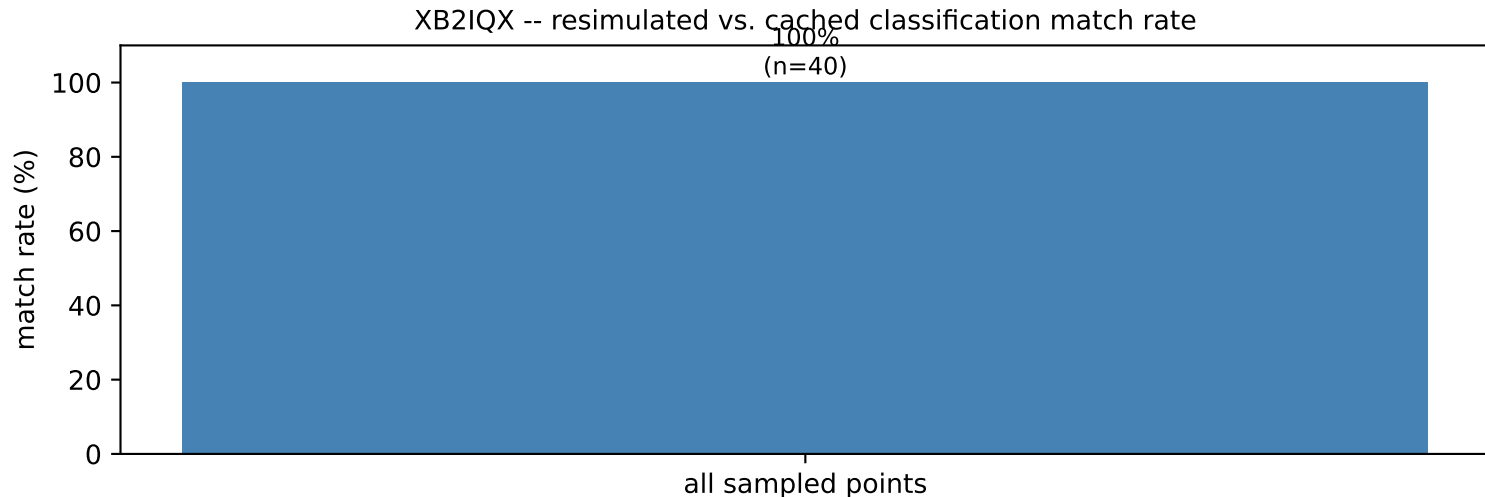
XB2IQX -- F-I relationship along held=0



39 non-silent, confidently-classified points along the held=0 column (compute\_fi\_slope's own filter -- flat zero-rate points below rheobase-equivalent excluded so they don't bias the linear fit). This scalar (fi\_slope\_hz\_per\_nA) already feeds the cross-cell feature table; this is simply the first time it's been plotted against the data it was fit to.

python src/generate\_defense\_packet.py --cells XB2IQX  
source: cell\_held\_injected\_grid.pkl -> cell\_grid\_features.pkl (extract\_grid\_features.compute\_fi\_slope())

## 10. Self-consistency check



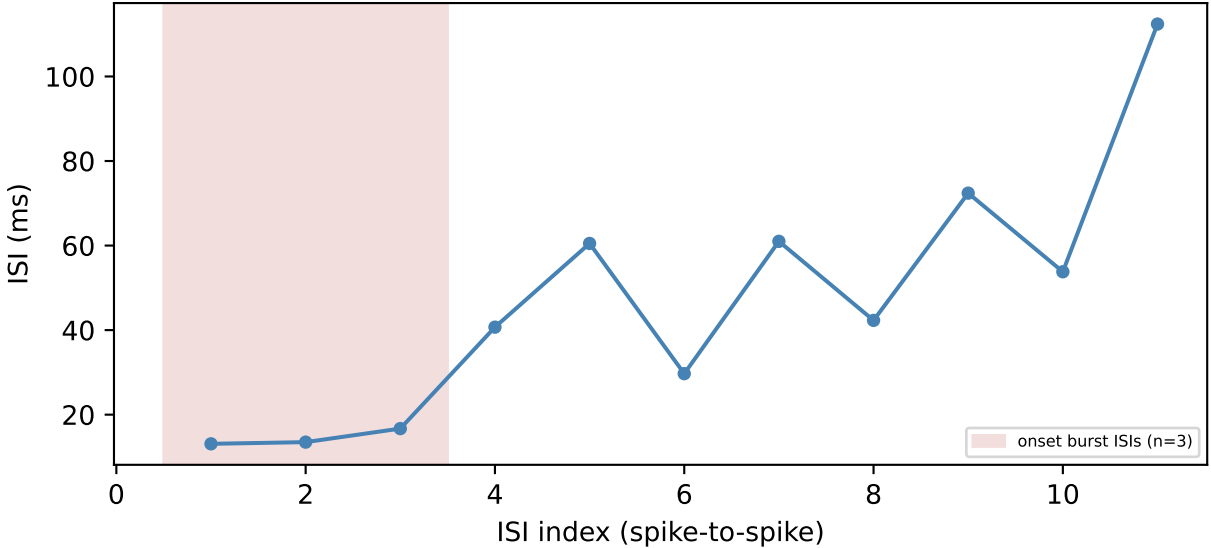
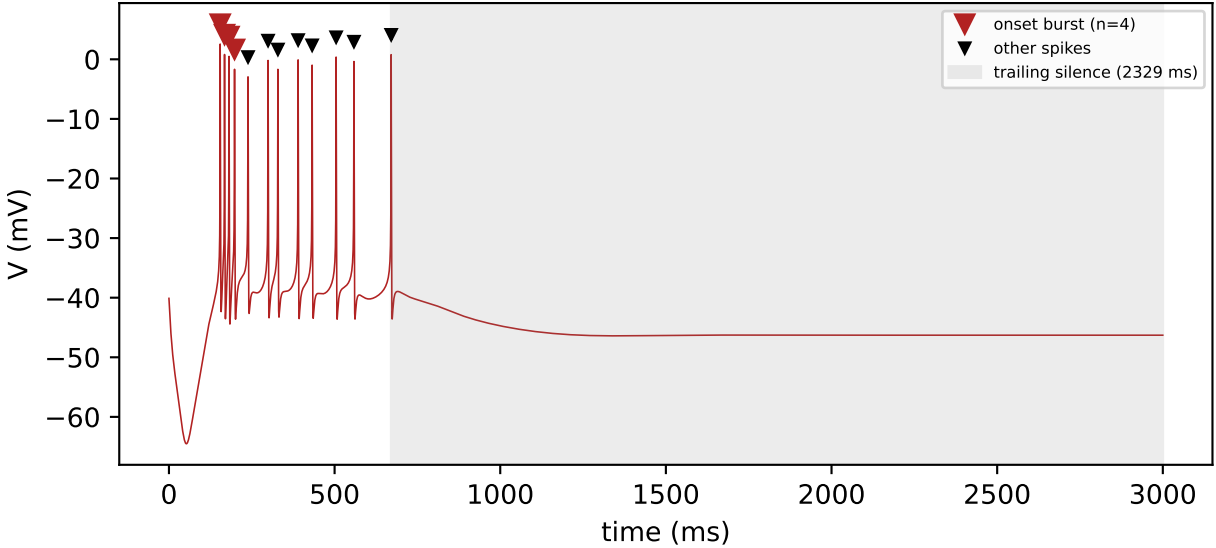
Re-simulates 40 random cached grid points end-to-end and checks whether the resulting test\_pattern/rebound\_pattern still matches what's stored in cell\_held\_injected\_grid.pkl. Match rate: 100%. Every held level in the current uniform-grid sweep (except held=0, which reuses the cell's own cached limit-cycle state exactly) is settled via a warm-start from whichever already-settled held level is numerically nearest -- the module docstring in run\_held\_injected\_grid.py already documents that this model shows path-dependent (hysteretic) settling near dynamical transitions, so a resimulation that warm-starts from a different already-settled state than the original sweep used can land on a different classification near a boundary. Any mismatches here are consistent with that

known property, not a new bug.

python src/generate\_defense\_packet.py --cells XB2IQX  
source: cell\_held\_injected\_grid.pkl -> cached test\_pattern/rebound\_pattern vs. fresh resimulate\_point() results

# 11. Burst-onset-then-silence detection

burst then ceased: held=-2.48 inj=-4.04 -> pattern='tonic', adaptation\_ratio=None (suppressed)



Onset burst detected: 4 spikes (mean ISI 14.4 ms) at window start, found by scanning locally for the first ISI jump -- independent of whatever whole-window tonic/bursting/silent label the KDE bimodality test assigns. Trailing silence after the last spike: 2329 ms of a 3000 ms window. likely\_ceased\_firing=True -- 2329 ms is  $\geq 3 \times$  the most recent ISI, so test\_adaptation\_ratio is suppressed (reported as None): a first-k/last-k ratio isn't a meaningful 'smooth adaptation' number for a train that stopped partway through the window.

burst then sustained: held=-3.77 inj=-3.37 -> pattern='tonic', adaptation\_ratio=9.51

